PROSPERO
GROUP
"the freedom to be unique"

KINGDOM ARTS ECOSYSTEM

Platform Architecture & Implementation Blueprint

Music School + Talent Agency Integration

Shared Backend Infrastructure with Separate User Experiences

March 2026

Executive Summary

The Kingdom Arts ecosystem comprises three interconnected components: a public marketing website, a music school application, and a talent agency application. This blueprint outlines a unified architecture where both the school and agency share a single backend infrastructure while maintaining distinct user experiences optimized for their respective audiences.

Key Strategic Decisions

- Two separate web applications (school.kingdomarts.org.bw & agency.kingdomarts.org.bw)
- One shared Supabase backend for authentication, profiles, and data
- Opt-in mechanism: school students can choose to become agency talent (not automatic)
- External musicians onboarded through a focused certification track in the school app before joining the agency
- Clean separation of UX and concerns; seamless backend integration

Architecture Overview

The proposed architecture uses a backend-as-a-service (BaaS) approach with Supabase as the central hub. Both applications authenticate against the same database, share user profiles, and maintain data consistency through a unified schema.

System Architecture Diagram

Component	Platform	Primary Purpose	Key Features
SHARED BACKEND INFRASTRUCTURE	Supabase	Centralized User Management & Core Services	• User accounts (one identity), • Profiles (student info, musician portfolio), • Credentials/Certifications (school completion) • Messaging & Payments (shared across both apps)
SCHOOL APP	school.kingdomarts	Educational Platform for Students	• Courses, • Lessons, • Live Sessions, • Practice Tools, • Certification, • [OPT-IN TOGGLE] "Join Agency"
AGENCY APP	agency.kingdomarts	Professional Management for Musicians & Clients	• Job Board, • Musician Profiles, • Auditions, • Bookings Management, • Client Directory, • Payment Processing

Three Key User Flows

Flow 1: KAA Student → Agency (Opt-in)

- Student reaches Level 5+ or completes a performance certification
- Dashboard shows: "You're eligible! Interested in paid gigs?"
- They click "Join Agency Talent Pool"
- Their profile + portfolio automatically syncs to the agency (with permission)
- They can now log into agency.kingdomarts.org.bw with the same credentials
- Agency sees them as "KAA Certified — Piano Level 7" (with credentials)

Flow 2: External Musician → School Onboarding → Agency

- External musician visits agency.kingdomarts.org.bw (no account yet)
- Sees: "Not certified yet? Complete our onboarding"
- Agency invites them to school.kingdomarts.org.bw for a Musician Certification Track (8-12 weeks)
- They take focused courses (not full school curriculum), get evaluated
- Upon completion, they're certified and auto-enrolled in agency
- Now they're verified talent in the system with credentials

Flow 3: Pure School Students (No Agency Involvement)

- Take courses, get better, graduate. Never interact with agency.
- Unaffected by talent trade. Privacy respected. Parents comfortable.
- Can change mind anytime and opt-in to agency (if they become eligible)

Shared Database Schema (High Level)

The Supabase database contains the following core tables. Both applications query and update these tables.

Table	Purpose	Used By	Key Fields
users	Auth + identity	Both apps	email, password, role
profiles	User details	Both apps	name, bio, avatar, location
musician_profiles	Portfolio + credentials	Both apps	instruments, genres, level, samples, visible_to_agency
certifications	Verify credentials	Both apps	cert_type, level, date_issued, expires
courses	School content	School app	title, instructor, level, duration
job_listings	Agency placements	Agency app	title, category, gig_type, dates, rate
bookings	Gig assignments	Agency app	musician_id, job_id, status, payment

PROSPERO
GROUP
"the freedom to be unique"

Feature Breakdown by Application

School App (school.kingdomarts.org.bw)

Student Features

- Browse and enrol in courses by instrument and genre
- Watch video lessons and download practice materials
- Practice tools (tuner, metronome, breath meter, etc.)
- Join live sessions and group ensembles
- Virtual stage: perform and be reviewed
- Track achievements and level progression
- Community: message instructors and peers
- [OPT-IN] When eligible: "Interested in paid gigs? Join our Talent Agency"

Instructor Features

- Create and upload courses and lessons
- Manage student enrollments and progress
- Grade assignments and provide feedback
- Host live sessions
- View royalty statements

Admin Features

- User management and role assignment
- Content approval queue
- Analytics dashboard
- Certification issuance

Agency App (agency.kingdomarts.org.bw)

Musician Features

- Build and manage musician profile (portfolio, instruments, genres, experience level)
- Upload audio/video samples and certifications
- Browse job board (Cruise Ships, Bands, Concerts/Festivals, etc.)
- Apply/audition for gigs
- View booked gigs and accept/decline offers
- Track payments and view earnings
- For non-certified externals: see the onboarding pathway to the certification track

Agency Admin Features

- Post and manage job listings
- Review musician applications/auditions
- Manage bookings and gig assignments
- Process payments and track musician earnings
- Invite external musicians to the onboarding in the school app
- Analytics: musician performance, gig placements

Technology Stack

Backend

- Database: Supabase PostgreSQL
- Authentication: Supabase Auth (JWT)
- File Storage: Supabase Storage (audio, video, documents)
- Real-time: Supabase Realtime (for live features)
- API: Supabase REST API + custom Edge Functions (if needed)

Frontend (Both Apps)

- Framework: React 18 with TypeScript
- Build Tool: Vite
- Styling: Tailwind CSS v4
- UI Components: Radix UI
- State Management: Zustand
- Icons: Lucide React
- Video: Jitsi Meet (embedded)
- Animations: Motion/Framer Motion

PROSPERO
GROUP
"the freedom to be unique"

Implementation Roadmap

Phase 1: Backend Setup & School App Backend Integration

Estimated: 1 week

- Design and create a Supabase database schema
- Set up authentication and user role management
- Integrate school app with Supabase (auth, courses, profiles)
- Wire up lesson uploads and live session infrastructure
- Test authentication flows and role-based access

Phase 2: Agency App Development

Estimated: 2-4 weeks

- Design and implement a musician profile system
- Build a job board and job listing management
- Implement musician application/audition system
- Build booking and contract management
- Integrate payment processing

Phase 3: Cross-App Integration & Onboarding

Estimated: 1 week

- Build an opt-in mechanism for school students to join the agency
- Create a certification track for external musicians in the school app
- Build a dashboard switcher to move between the school and the agency
- Test end-to-end flows for all user types

Phase 4: Deployment & Launch

Estimated: 1-2 weeks

- Deploy the school app to school.kingdomarts.org.bw
- Deploy agency app to agency.kingdomarts.org.bw
- Security audit and penetration testing
- Load testing and performance optimisation
- Launch and monitor for issues

PROSPERO BRENT. **Plot** 726, Unit 1, Mogoditshane, Gaborone, Botswana, Po Box 1338, Mogoditshane, **Tel**: +267 75209597,
Cell: +267 71881185, **Email**: pkwelagobe@gmail.com

Operational Decisions Needed from Client

The following questions need to be answered before detailed implementation begins:

1. Student Eligibility & Opt-in Mechanism

- At what level/achievement does a student become eligible to opt-in? (e.g., Level 5+, completed performance certification, teacher recommendation)
- Should students be notified proactively when they are eligible, or should they discover it on their own?
- Can students opt-out after joining the agency talent pool?

2. External Musician Onboarding

- What is the certification track for external musicians? (e.g., full school curriculum vs. condensed 8-week bootcamp, or workshops)
- Who pays for external musician onboarding? (school tuition, agency, or shared, royalty)
- Should external musicians be required to complete onboarding, or can certified external musicians be exempt from it?

3. Cross-App Visibility

- Once a student opts-in to the agency, should they see gigs/bookings in the school app dashboard, or only in the agency app?
- Should school admins see who has opted-in to the agency?

4. Payment & Revenue Model

- What is the revenue split between Kingdom Arts and musicians for gig payments?
- Do school course enrollments have different pricing models? (e.g., one-time purchase, subscription, free trial)

Next Steps

- Review this blueprint with stakeholders and confirm architectural decisions
- Answer the operational questions above (section: Decisions Needed from Client)
- Finalize timelines and resource allocation for each phase
- Begin Phase 1: Backend setup and database design
- Assign roles: who will manage content, oversee quality, handle admin?

This blueprint is ready for client presentation and discussion. Adjustments can be made based on feedback before implementation begins.

PROSPERO BRENT. **Plot** 726, Unit 1, Mogoditshane, Gaborone, Botswana, Po Box 1338, Mogoditshane, **Tel**: +267 75209597,

Cell: +267 71881185, **Email**: pkwelagobe@gmail.com